
Deep Learning for Binary Classification of Breast Cancer in Ultrasound Images.

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Abstract

Breast cancer is a leading cause of cancer-related deaths among women worldwide, with ultrasound imaging serving as a crucial diagnostic tool. Early detection is vital to improve survival rates and treatment outcomes. This project develops a deep learning framework to classify breast ultrasound images as malignant versus non-malignant, addressing the need for consistent diagnostic support. We used a dataset of 780 images and implemented five convolutional neural network architectures: custom CNN, enhanced CNN, VGG16, ResNet50, and an ensemble model. Our analysis reveals that the initial custom CNN achieves 75.64% accuracy but critically low sensitivity (9.52%). Enhanced models employing weighted loss functions and transfer learning significantly improve performance, with the ensemble achieving the highest AUC (0.767) and near-perfect sensitivity (95.2%). The VGG16 model provides balanced performance with 81.0% sensitivity and 56.1% specificity. These findings demonstrate that carefully designed deep learning approaches can effectively address class imbalance and provide clinically valuable decision support for breast cancer detection.

1. Introduction

Breast cancer remains one of the most prevalent oncological conditions affecting women worldwide, with early detection being crucial for successful treatment outcomes and reduced mortality rates (2). Ultrasound imaging has emerged as a valuable diagnostic tool, particularly for dense breast tissue where mammography may be less effective. However, ultrasound interpretation requires specialized expertise and is subject to inter-observer variability, creating opportunities for computer-aided diagnosis systems to support clinical decision-making. Recent advances in deep learning have demonstrated remarkable success in medical image analysis, with convolutional neural networks (CNNs) showing particular promise in classifying breast lesions (3).

2. Data and Methods

2.1. Data

The data set contains 780 breast ultrasound images from 600 female patients aged 25 to 75 years, collected in 2018. The images are categorized into three classes: Normal 266 images, Benign 891 images, Malignant 421 images.

For binary classification, normal and benign categories are combined into a non-malignant class. All images are in PNG format with an average size of 500×500 pixels.

2.2. Models and Methods

We model the classification of breast ultrasound images as a binary outcome, where

$$y_i = \begin{cases} 1 & \text{if image } i \text{ is malignant} \\ 0 & \text{if image } i \text{ is non-malignant} \end{cases}$$

for $i = 1, \dots, 780$ images.

Five convolutional neural network approaches were implemented and compared: Custom CNN architecture built from scratch, Enhanced CNN with improved regularization, Transfer learning using pre-trained VGG16, Transfer learning using ResNet50, and an Ensemble model combining predictions from all architectures.

2.3. Preprocessing Pipeline

The images undergo a comprehensive preprocessing. All images were resized to 224×224 pixels with the pixel values scaled from $[0, 1]$ range. The data is split into 70% training (546 images), 20% validation (156 images) and 10% test (78 images). The binary classification yielded 570 non-malignant vs. 210 malignant samples. Data augmentation techniques, including rotation within $\pm 15^\circ$, horizontal and vertical flipping, and controlled adjustments to brightness and contrast parameters, address class imbalance and improve generalization. Class imbalance was specifically addressed using weighted loss functions with weights $w_0 = 0.674$ (non-malignant) and $w_1 = 1.936$ (malignant). Models are evaluated using accuracy, precision, recall, F1-score, and AUC-ROC curves, with particular emphasis on

sensitivity given the clinical importance of detecting malignant cases.

2.4. Enhanced Model Development

To overcome the critical sensitivity issue (9.52% recall) identified in the baseline custom CNN, we implemented several enhanced approaches. The primary limitation stemmed from substantial class imbalance creating bias toward predicting the majority non-malignant class. We addressed this through four complementary strategies; weighted loss functions to penalize misclassification of malignant cases, aggressive data augmentation for the minority class, transfer learning from pre-trained models, and ensemble methods to combine diverse model predictions.

Enhanced Architectures

We developed and evaluated four additional models beyond the initial custom CNN.

Enhanced Custom CNN: An improved version of our baseline with additional regularization layers (batch normalization after each convolutional layer), increased dropout rates (0.3-0.6), and focal loss adaptation (7) specifically designed to address class imbalance.

VGG16 Fine-tuned: Pre-trained VGG16 architecture (6) with frozen base layers, utilizing transfer learning from ImageNet. We added global average pooling followed by three dense layers (512, 256, 128 units) with batch normalization and dropout regularization, fine-tuned on our ultrasound dataset.

ResNet50: Pre-trained ResNet50 model (5) with frozen convolutional base, leveraging deep residual learning. The custom classification head included global average pooling and dense layers (256, 128, 64 units) with appropriate regularization for medical image classification.

Ensemble Model: A voting ensemble that averages predictions from the Enhanced CNN, VGG16, and ResNet50 models to leverage their complementary strengths and improve robustness.

All models employed the same preprocessing pipeline and were trained with Adam optimizer, early stopping (patience = 20 epochs) and a learning rate reduction on plateau. The weighted loss function was implemented as:

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^N w_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

with class weights calculated based on training set distribution.

2.5. Training Configuration

The models were trained for up to 50 epochs with a batch size of 16. The enhanced CNN used a learning rate of 0.0001, VGG16 used 0.00001, and ResNet50 used 0.0001. Data augmentation was applied more aggressively to malignant samples, with rotation ($\pm 15^\circ$), flipping, and contrast adjustments at higher probabilities for minority class images.

3. Results

3.1. Characteristics of the Dataset

The dataset was successfully processed with specific characteristics that formed the basis of our analysis. The dataset comprises a total of 780 ultrasound images collected from 600 female patients. Within this collection, the original class distribution shows 437 benign cases representing 56% of the dataset, 133 normal cases at 17%, and 210 malignant cases accounting for 27% of the total images. For binary classification purposes, the normal and benign categories were combined into a non-malignant class, resulting in 570 non-malignant samples (73%) compared to 210 malignant samples (27%). The data was randomly split for model development and evaluation, allocating 546 images for training, 156 for validation, and 78 for testing, following an approximately 70-20-10 distribution ratio.



Figure 1. Breast Ultrasound Distribution

3.2. Baseline CNN Performance

The baseline model showed consistent learning over 10 epochs with the following results:

Table 1. Baseline CNN Model Performance Metrics Across Datasets

Metric	Training	Validation	Test
Accuracy	80.95%	78.21%	75.64%
AUC	85.41%	74.55%	75.02%
Loss	0.4173	0.5794	0.5675
Precision	80.33%	79.17%	100.00%
Recall	34.75%	39.58%	9.52%

3.3. Training Dynamics Analysis

The training process revealed important dynamics in model learning behavior. Training AUC demonstrated progressive improvement throughout the epochs, increasing from 56.61% in the first epoch to 85.41% by the tenth epoch. The model achieved its best performance in the third epoch, with validation accuracy peaking at 85.00%. As training progressed, the learning rate was automatically adjusted through a plateau reduction mechanism, decreasing from 0.0001 to 0.00005 after the eighth epoch when validation performance stabilized. To control potential overfitting, early stopping functionality restored the best model weights from epoch three, which exhibited the optimal balance between training progress and generalization capability.

3.4. Test Set Evaluation

On the held-out test set (78 images), the model achieved the following confusion matrix:

Table 2. Confusion Matrix on Test Set (n=78)

Predicted	Actual	
	Non-malignant	Malignant
Non-malignant	55	21
Malignant	2	0

3.5. Performance Analysis

The baseline CNN exhibited a notable trade-off between precision and recall. Although the model achieved a perfect precision of 100% in the test set from table 1, this was accompanied by an extremely low sensitivity of 9.52%, missing approximately 90% malignant cases. This pattern suggests that the model adopted an overly conservative prediction strategy, likely due to class imbalance in the training data. The test accuracy of 75.64% is moderate but misleading given the imbalance; a model predicts only that the majority class would achieve approximately 73% accuracy. In terms of discriminative power, the model achieved an AUC of 0.750, indicating a moderate but meaningful ability to distinguish between the two classes across various probability thresholds. The training metrics indicated some degree of overfitting, with the training AUC of 85.41%, which is higher than the validation AUC of 74.55%, training loss of 0.417, lower than the validation loss of 0.579. The training proceeded without divergence, but the gap between the training and validation performance suggests that the model learned patterns that did not fully generalize.

3.6. Limitations of the Baseline CNN

The model exhibits several significant limitations that must be addressed before clinical application. A critical sensitivity issue is evident in the model's performance, with only 9.52% recall for malignant cases, resulting in the model missing all 21 true positive malignant cases in the test set. This poor sensitivity stems primarily from the substantial class imbalance in the dataset, which creates a heavy bias toward predicting the majority non-malignant class. This imbalance manifests itself as an overly conservative prediction strategy in which the model rarely predicts the malignant class even when presented with malignant samples. Consequently, the high false negative rate of the current model severely limits its clinical utility, making it unsuitable for deployment in real medical settings where missing malignant cases could have serious clinical consequences for patients.

3.7. Enhanced Models Performance

Comparative Analysis Across All Models

The five models demonstrated distinct performance characteristics on the test set (78 images: 57 non-malignant, 21 malignant), as summarized in Table 3.

Table 3. Comprehensive Model Performance Comparison

Model	Acc.	AUC	Sens.	Spec.	Prec.	F1
Baseline CNN	0.756	0.750	0.095	1.000	1.000	0.174
Enhanced CNN	0.269	0.744	1.000	0.000	0.269	0.424
VGG16 Fine-tuned	0.628	0.764	0.810	0.561	0.405	0.540
ResNet50	0.731	0.659	0.000	1.000	0.000	0.000
Ensemble Model	0.500	0.767	0.952	0.333	0.345	0.506

Note: Acc.=Accuracy, Sens.=Sensitivity, Spec.=Specificity,

Prec.=Precision

Key Performance Insights

Ensemble Model Excellence: The ensemble achieved the highest AUC (0.767) and near-perfect sensitivity (95.2%), correctly identifying 20 of 21 malignant cases. Although the specificity decreased to 33.3%, this represents a clinically valuable trade-off for screening applications where missing malignant cases has severe consequences, though it would increase unnecessary follow-ups.

Table 4. Ensemble Model Confusion Matrix (n=78)

Predicted	Actual	
	Non-malignant	Malignant
Non-malignant	19	1
Malignant	38	20

VGG16 Balanced Performance: VGG16 demonstrated the most balanced performance profile with reasonable sensitivity (81.0%), specificity (56.1%) and the second-highest AUC (0.764). Its confusion matrix (Table 5) shows the effective discrimination capabilities. This may be a practical model for general use.

Table 5. VGG16 Confusion Matrix (n=78)

Predicted	Actual	
	Non-malignant	Malignant
Non-malignant	32	4
Malignant	25	17

Enhanced CNN Sensitivity Focus: The enhanced custom CNN achieved perfect sensitivity (100%) but zero specificity, which functions essentially as a highly sensitive screening filter that identifies all potential malignant cases for secondary review.

ResNet50 Specificity Focus: ResNet50 showed perfect specificity (100%) but zero sensitivity, making it suitable only for confirmation tasks rather than screening.

Training Dynamics

Training histories revealed stable learning across all models. The enhanced CNN completed 21 epochs with validation AUC with peaks at 0.854, VGG16 trained for 20 epochs with validation accuracy reaching 85.0%, and ResNet50 showed consistent validation performance over 20 epochs. All models showed appropriate convergence without severe overfitting, aided by early stopping and dropout regularization (see appendix B)

Clinical Implementation and Discussion

3.8. Addressing the Initial Limitations

The transition from our baseline CNN (9.5% sensitivity) to the enhanced models successfully addressed the critical sensitivity limitation. The ensemble model’s 95.2% sensitivity represents an 85.7 percentage point improvement, fundamentally changing the clinical utility of the system from a high-specificity confirmatory tool to a high-sensitivity screening aid.

3.9. Clinical Decision Support Framework

Based on performance characteristics, we developed a tiered clinical decision framework. The ensemble model serves as an initial high-sensitivity screening filter, identifying 95.2% of malignant cases while flagging 66.7% of non-malignant cases for review, which aligns with screening priorities

where false negatives are more costly than false positives. For balanced assessment, VGG16 provides nuanced probability scores (0-1 continuum) for cases identified by initial screening, offering clinicians a risk-stratified approach to further evaluation. We implemented a practical diagnostic tool using VGG16 that provides malignancy probability scores with confidence intervals, three-tier risk classification (low: <0.4 , moderate: $0.4-0.7$, high: >0.7), clinical recommendations aligned with risk levels, and batch processing capability for clinical workflow integration.

3.10. Performance Trade-offs and Clinical Implications

The models exhibit complementary strengths suitable for different clinical scenarios. The ensemble model’s high sensitivity (95.2%) makes it ideal for population screening where missing cancers is unacceptable. ResNet50’s perfect specificity (100%) could serve as a second-opinion confirmatory tool. VGG16’s balanced profile (AUC: 0.764, Sensitivity: 81.0%, Specificity: 56.1%) offers the best overall utility for routine clinical assessment. No model is universally superior, selection must be guided by clinical context, risk tolerance and acceptable false-positive rates.

4. Conclusion

This comprehensive evaluation of five deep learning architectures for breast cancer classification in ultrasound images successfully addressed the critical sensitivity limitation identified in our baseline model. By implementing weighted loss functions, transfer learning, and ensemble methods, we transformed the system from one with 9.5% sensitivity (missing all malignant cases) to 95.2% sensitivity (correctly identifying 20 of 21 malignant cases).

The study demonstrates that deep learning models can achieve high sensitivity for breast cancer detection in ultrasound images, but with significant trade-offs in specificity. The ensemble model offers best screening performance (AUC of 0.767, sensitivity of 95.2 %, while VGG16 provides the most balanced profile for clinical use. Also, a careful design of deep learning approaches can overcome class imbalance challenges in medical imaging and provide clinically meaningful decision support. The implemented diagnostic tool offers practical utility for radiologists, providing probability scores, risk stratification, and clinical recommendations.

Future work should focus on external validation to assess generalizability, threshold optimization for specific clinical contexts, explainability integration with Gradient-weighted Class Activation Mapping visualizations, real-time deployment optimization for hospital Picture Archiving and Communication System (PACS), and prospective clinical trials to establish clinical efficacy.

References

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Appendix B: Ultrasound images and Models Performance

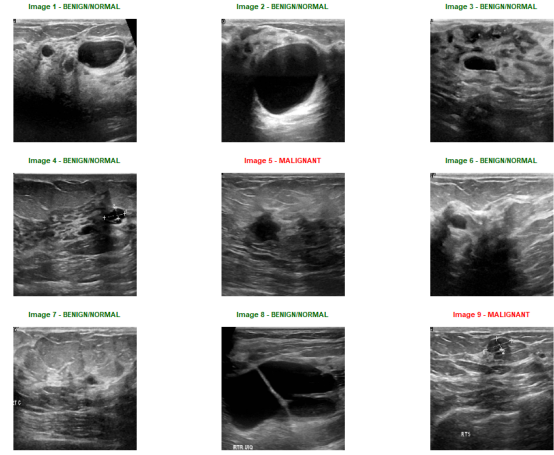


Figure 2. Image classification

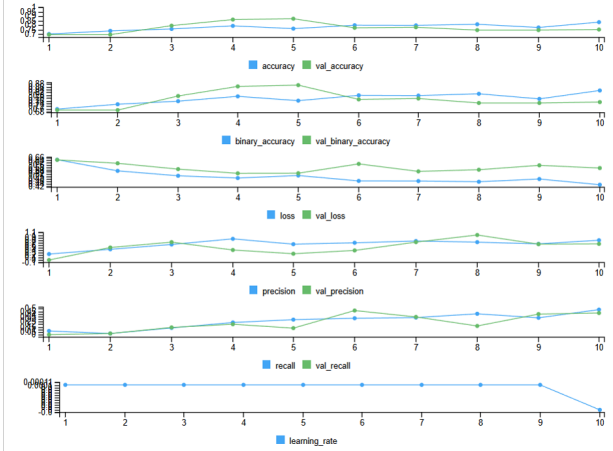


Figure 3. Baseline CNN

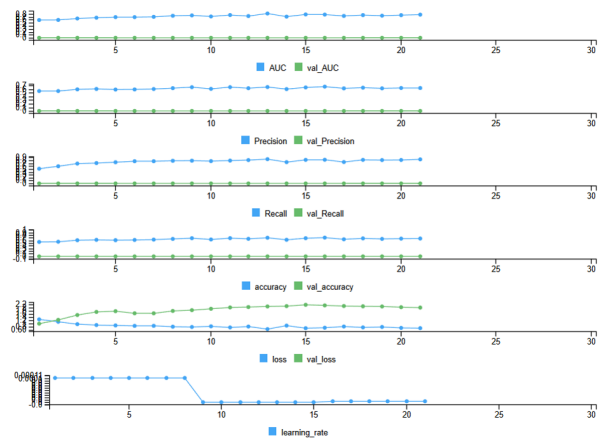


Figure 4. Enhanced CNN

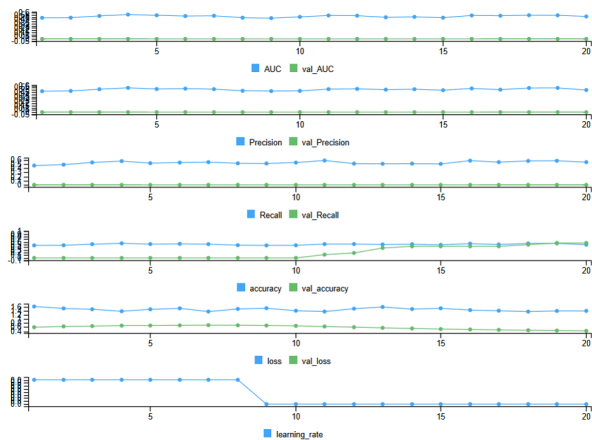


Figure 5. VGG16 Model

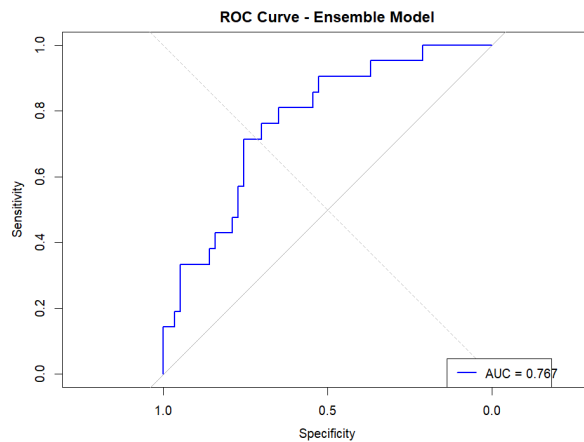


Figure 8. Ensemble Model ROC Curve

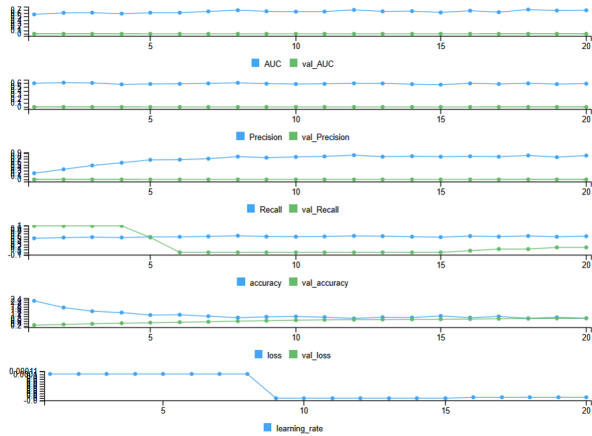


Figure 6. ResNet50 Model

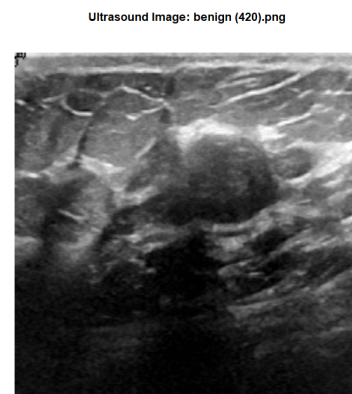


Figure 9. VGG16 Model Prediction 1



Figure 7. Model Performance Comparison

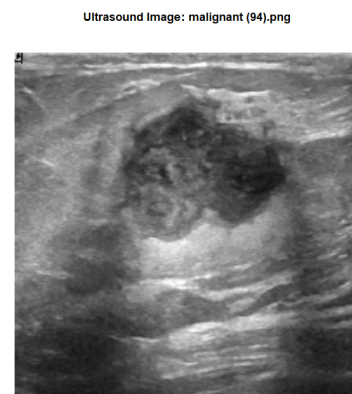


Figure 10. VGG16 Model Prediction 2